ENGINEERING PORTFOLIO · 2026

Ryan Dolan Rodriguez

Mechanical Engineering · University of Notre Dame · Class of 2028

From **Miami, Florida**.

SEEKING SUMMER 2027 INTERNSHIPS

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1 IRIS balloon payload
2 Spec retrieval & citation system
3 Rube Goldberg machine
4 Vibration isolation
5 Skills and experience



Ryan Dolan Rodriguez · Miami, FL

IRIS: A balloon payload that flies itself home

IRIS is a **5.53 lb** box that rides a weather balloon to about **89,000 ft**, lets go, drops on a parachute, then switches to a steerable parafoil and flies to a landing spot the team picks. I own structures design, testing, validation and fabrication on it, from the internal shelving to the waterjet-cut chassis, plus the power wiring.

OBJECTIVE

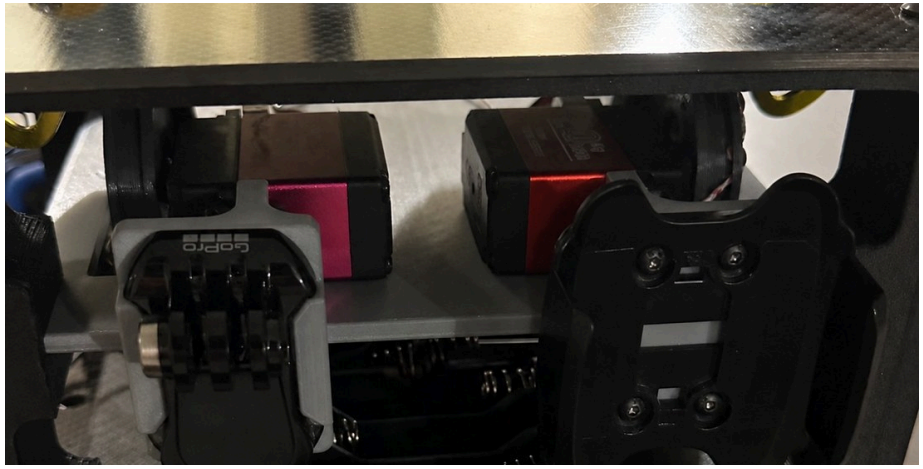
- Land the payload where we pick, so it can fly again.
- 5.53 lb inside 13.5 × 8 × 7.1 in, ~$1,200 a unit.
- Survive 70 min of climb, air far below freezing, and two deployment snaps.
- Rebuild between flights without a teardown. Parts made on campus.

OUTCOME & MY CONTRIBUTION

- **Flown to 89,294 ft**, plus **30+ drop tests** from 300 ft.
- **Designed the internal shelving, then rebuilt it.** Shelves lift out as modules, so a servo, camera or flight computer swaps in minutes, not a teardown. The current version adds a bottom and a middle shelf that route power and signal separately, which killed the wire tangling and keeps battery mass low for a steadier hang.
- **Ran CFD** on the deployment mechanism, cut drag uncertainty **30%+**, then sized it in Ansys.
- **Machined** the 6061 chassis plates on the waterjet.
- **Wired 3 isolated circuits**, then ran the cold, structural and electrical test plan. The box held **2.5+ h at −10 °C**.

TECHNICAL DETAILS & SKILLS

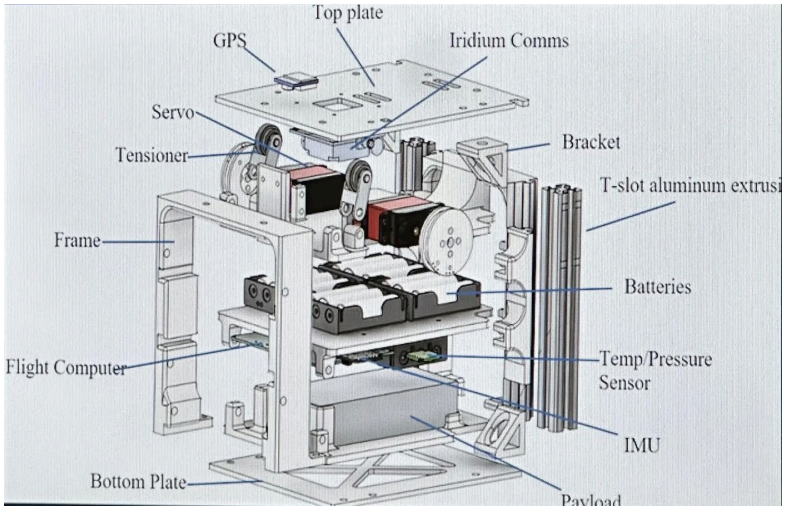
`SolidWorks · Ansys FEA and CFD · 6061-T6, carbon plate, printed shelving · waterjet and CNC mill · cold chamber testing`



The shelving I designed, current build. Bottom and middle shelves carry the battery pack and flight computer, servos and the camera mount bolt to fixed points above, and power and signal run on separate levels. The shelves double as stiffeners between the side panels, and everything lifts out as a module.



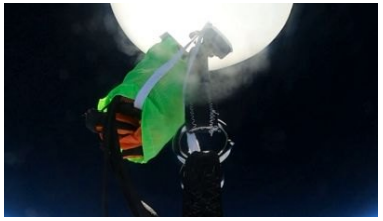
Flight configuration. Foam over the chassis for the cold, chute stowed on top, pinned bridle and D-rings for the launch crew.



Exploded CAD. Top and bottom plates on a T-slot aluminum frame, servos and tensioners for parafoil steering, and stacked shelves for the batteries, flight computer, IMU and temp/pressure sensor.



Parafoil descent. A dual 3-ring release swaps the parachute for the parafoil, then two servos steer it.



89,294 ft. Balloon still attached, parafoil stowed in green.

Engineering Specification Retrieval & Citation System

RETRIEVAL-AUGMENTED GENERATION · OCR · CITATION TRACING

Built during an **AI Leaders of Tomorrow** fellowship through iNDustry Labs at Notre Dame for a **$1.3B** parts supplier. The system answers engineering-specification questions in seconds and cites the source document, section and page. Measured performance projected to an **80% reduction in lookup time** — approximately **3,000 engineering hours and $200K+** in annual capacity across 60–70 engineers.

OBJECTIVE

- Replace manual cross-referencing across hundreds of engineering specifications.
- Recover the **42%** of the library stored as unsearchable scanned images.
- Return every answer with its document, section and page, allowing an engineer to verify the source directly.
- Fail safely: on compliance-sensitive specifications, an unsupported answer is worse than returning no answer.

MY CONTRIBUTION

- **Built the pipeline end to end**: document ingestion → OCR → retrieval → answer synthesis → citation tracing.
- **Recovered the image-only portion** of the corpus, making 42% of the library searchable for the first time while improving scanned-document throughput **6×**.
- **Designed and graded a 103-question evaluation battery** against source documents: 86/103 correct and 0 unsupported responses.
- Built the impact model translating measured lookup performance into estimated engineering hours and capacity.
- Demoed the working system to the client's engineers and my program mentors.

TECHNICAL DETAILS & SKILLS

Python · Retrieval-Augmented Generation · OCR · Embeddings · Passage Retrieval · LLM APIs · Evaluation Design · Citation Tracing · Technical Documentation

MEASURED PERFORMANCE MEASURED



103-question evaluation. Graded question-by-question against the source document. Every returned answer resolved to a real document, section and page — a traceability result, not proof the system cannot return an incorrect interpretation.

42%
previously image-only library made searchable

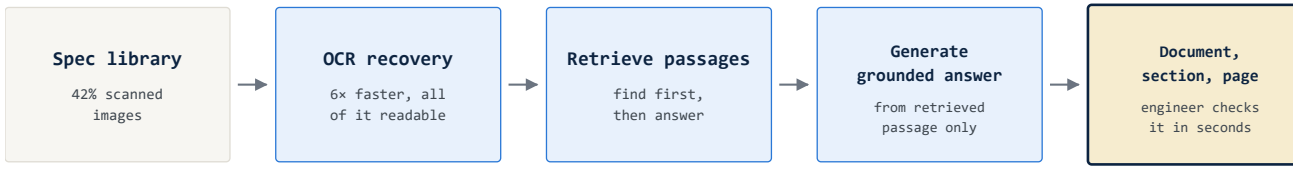
6×
scanned-document processing throughput vs. baseline

MODELED IMPACT · BASED ON MEASURED TOOL PERFORMANCE MODELED

$200K+
modeled annual engineering capacity

~3,000 h
modeled annual capacity · 60–70 engineers

80%
projected reduction in specification lookup time



Blue stages are the pipeline I built. Every returned answer resolves to a document, section and page so an engineer can verify it directly. **Under NDA**, so no company name and no documents here, only totals and method.

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Rube Goldberg Machine Project

A machine that shuts off a phone alarm. One push starts it, gravity does the rest, and a stylus taps the screen at the end. **32 parts** in **6 subassemblies**, **30 released drawings**, and it came in under budget. It sets up in about **15 seconds** against a 30-second target and shut the alarm off **every run**.

OBJECTIVE

- Do one simple thing in as many steps as possible. 8 in a row, under 4 min.
- **Every part machinable.** No wood, no glue, no printed parts. That rule drove the build.
- Set up in 30 s or less, inside a 24 × 36 in footprint, under a $300 cap.
- Gravity only after the first push.

OUTCOME & MY CONTRIBUTION

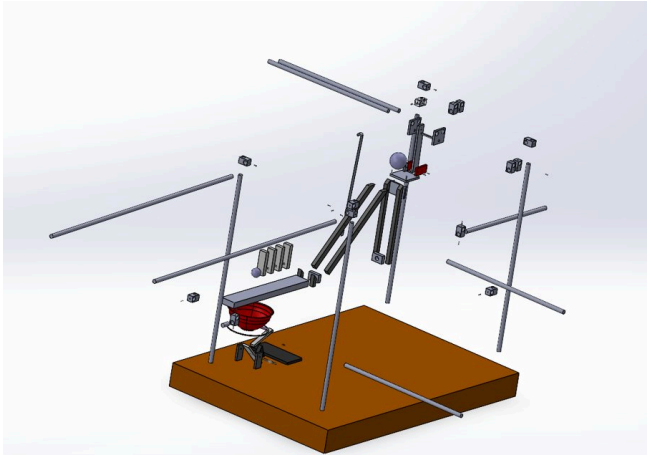
- **Owned design for assembly and manufacturability.** I reviewed every part against the tools we had before release, and drove the five changes that came out of it.
- **Designed** the slide and the domino platform, 9 of the 32 parts, then cut, drilled, tapped and filed them myself.
- **Cut setup to about 15 seconds**, half the target, by fixing how the subassemblies locate against each other.
- **Sourced the material** for the whole machine and brought it in under budget.
- Alarm went off on **every run**, start to finish in about 5 seconds.

TECHNICAL DETAILS & SKILLS

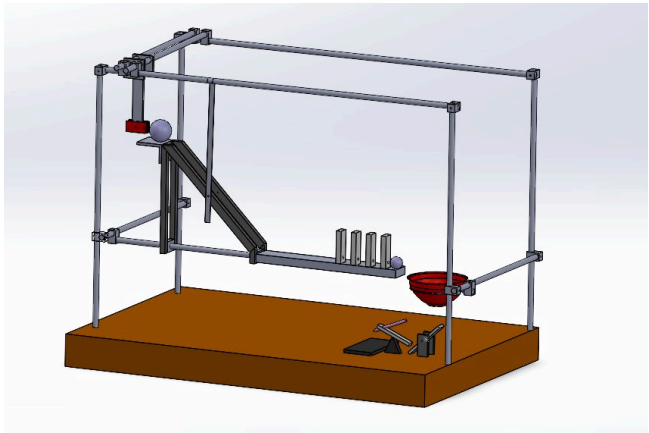
SolidWorks · design for assembly and manufacturing · ASME Y14.5 and GD&T · 0.25 and 0.5 in 6061 plate, bar and rod · bandsaw, drill press, hand tapping, filing to fit · sourcing and bills of materials



The built machine. My slide and domino platform run down the right side. Ball leaves the slide, tips 4 dominoes, drops through the funnel, hits the seesaw, and the lever pushes the stylus onto the phone.



Final exploded view. All 32 parts across 6 subassemblies, numbered against the bill of materials.



Final CAD assembly. Every difference between this and the photo is one of the five changes at bottom right.

Drawn in CAD → built in the shop. The five calls I made once the tools were the constraint.

PART	CHANGE
Pendulum	6 pieces → 3. Extra pieces did nothing.
Slide	Compound angles → simple cuts, screwed together.
Domino platform	Aluminum → foam board with aluminum stiffeners and rails. Too heavy, then it spun on its rod.
Seesaw lever	Acrylic → aluminum. Acrylic could not push the next step.
Lever base	Triangle → square with foam feet. Small angled cuts are slow to clamp.

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Vibration Isolation Project

A rocket shakes hard on the way up and the satellite bolted to it feels all of that. I drove the shake input straight from **SpaceX's published Falcon 9 User's Guide** and sized rubber isolators for a **2,982 kg** satellite from 5 to 100 Hz, validated the model three ways in MATLAB and Python.

OBJECTIVE

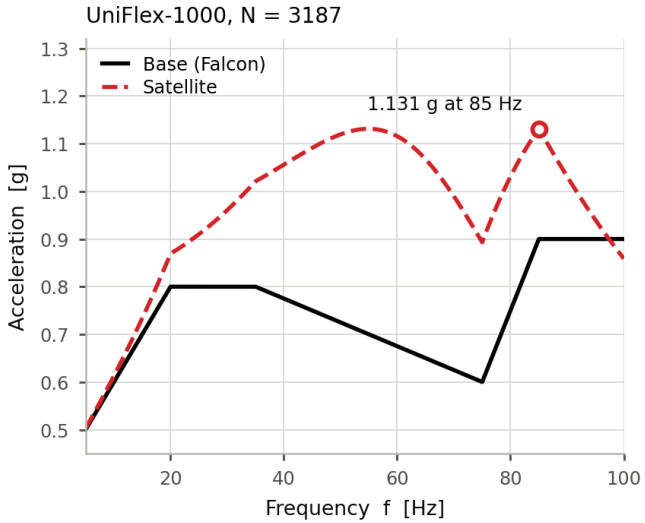
- Pick the isolator and the count that leaves the satellite shaken as little as possible from 5 to 100 Hz.
- Keep the isolated frequency above 25 Hz, and use 4 isolators or more.
- Shake input taken from **SpaceX's published Falcon 9 User's Guide**, 0.5 to 0.9 g.
- Stiffness from Moog's SoftRide datasheet, 1,000 / 3,000 / 6,500 lb/in.

OUTCOME & MY CONTRIBUTION

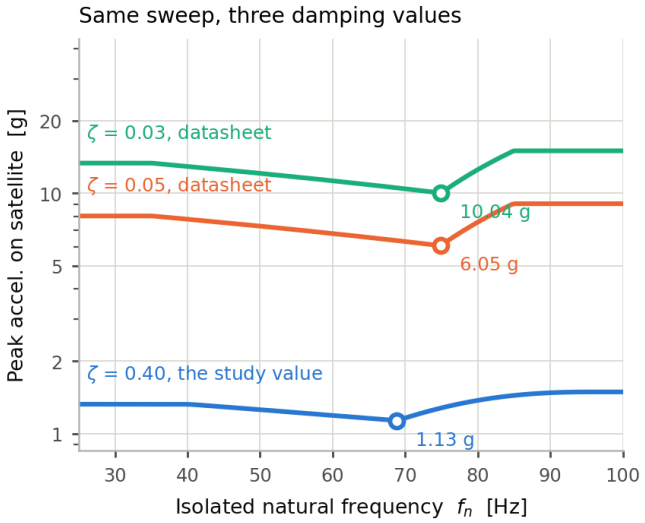
- **Validated** the model 3 ways first. Steady push settles at −6.00 mm and 180 N, sine shake lands within **0.12%** of the hand solution, backwards check to 0.37 mm.
- **Swept** every legal combination: UniFlex-1000, **1.131 g** worst case, 68.85 Hz isolated. The **3,187** isolators it wants are what the 25 Hz floor forces under 2,982 kg, and I flagged that rather than ship the number.
- **Showed damping runs the answer.** The project said 0.40, the datasheet says 3–5%, and the load moves from **1.13 g to 6–10 g** on that one assumption.

TECHNICAL DETAILS & SKILLS

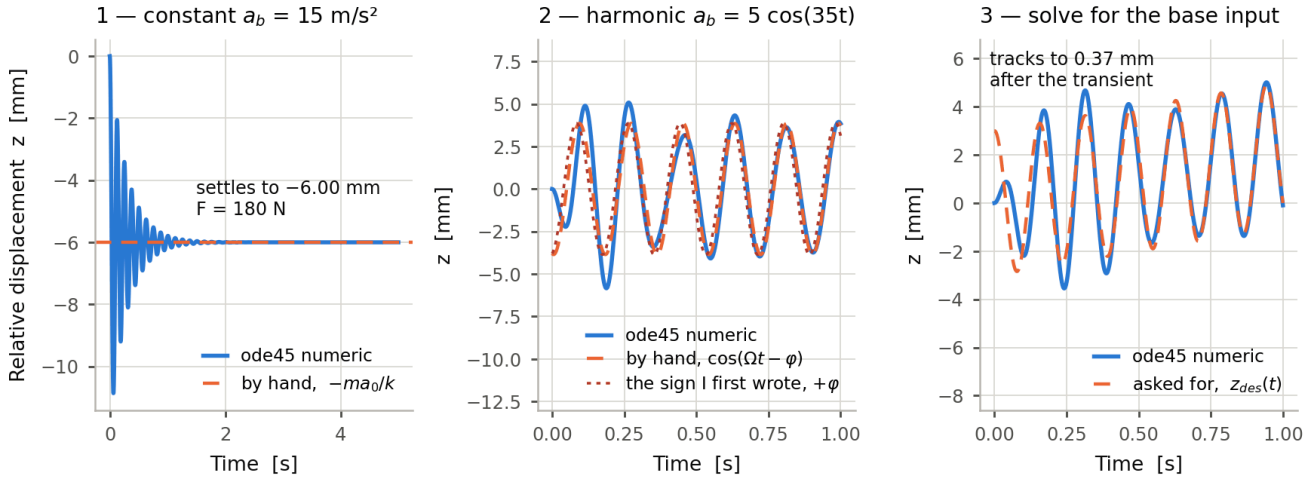
Base-excited spring, mass and damper · MATLAB with ode45 · Python, NumPy, SciPy · model validation · sensitivity study



What my script prints for the winning setup. Black is the SpaceX Falcon 9 input, red dashed is the satellite, from $T(r) = \sqrt{(1+(2\zeta r)^2)}/\sqrt{((1-r^2)^2+(2\zeta r)^2)}$. Worst case 1.131 g at 85 Hz, isolated frequency 68.85 Hz.



One assumption moves the load 5 to 9 times. Only the damping number changed between these three curves. This is the number I would argue about on a real program.



The three checks I ran first, from falconProjectPart1.m. m = 12 kg, c = 80 N·s/m, k = 30 kN/m. The middle panel shows both signs, the corrected one sitting on the numbers and the version I first used running 20.7° out of step.

Skills

Everything here is backed by the four projects on the pages before this one.

DESIGN AND SHOP

CAD and drawings
SolidWorks, Fusion 360, GD&T, drawings to ASME Y14.5

Machining and fabrication
Waterjet, CNC mill, bandsaw, drill press, hand tapping, filing to fit, 3D printing, some injection molding

Design for assembly and manufacturing
Picking stock and process before modeling, manufacturability reviews, sourcing, bills of materials

Languages
English and Spanish, fluent

ANALYSIS, TEST AND CODE

FEA and CFD
Ansys for stress and deflection, plus CFD on deployment hardware

Modeling and validation
MATLAB with ode45, Python with NumPy and SciPy, checking a model before trusting it

Hardware testing
Cold chamber runs, test plans for structure and electronics, battery budgets, design reviews

Technical writing and reviews
Test plans, drawing checks, design reviews, report sections

Code and AI systems
Python, MATLAB, retrieval augmented generation, OCR pipelines, LLM APIs, LaTeX

EXPERIENCE AND EDUCATION

MAY – AUG 2026

AI Leaders of Tomorrow Intern, iNDustry Labs at Notre Dame

Built a retrieval augmented generation spec synthesis agent for a $1.3B parts supplier. Modeled at $200K+ and ~3,000 engineering hours a year. 83.5% correct, 0 unsupported answers.

AUG 2025 – NOW

IRIS team, IrishSat, Notre Dame

Own structures design, testing, validation and fabrication on the balloon payload, plus the power wiring. CFD and FEA, waterjet-cut chassis, 3 isolated circuits, the test plan.

2025 – 2026

Conference Chair, Society of Hispanic Professional Engineers, Notre Dame

Ran a 40+ person trip to the national conference and raised $10,000+ at gameday fundraisers to cover travel and registration.

SUMMER 2025

Study abroad, Universidad Politécnica de Valencia, Alcoy, Spain

A semester of engineering classes in Spanish.

2024 – 2028

University of Notre Dame

B.S. Mechanical Engineering, minor in Engineering Corporate Practice.

Happy to talk through any of this.
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IRIS photos: IrishSat. CAD and drawings from the AME 21267 project. Vibration plots come from my own scripts, falconProjectPart1.m and isolatorProject.m, using Moog's published datasheet and the published Falcon User's Guide. Internship work is shown in totals only, under NDA.